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COMPLETE SPECIFICATION

(See section 10 and M.13)

SMART BICYCLE PROTECTIVE HELMET WITH SELF-HEALING CAPABILITIES USING
DIELS ALDER RESIN

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The following specification particularly describes the invention and the manner in which it is to be performed

SMART BICYCLE PROTECTIVE HELMET WITH SELF-HEALING CAPABILITIES USING DIELS ALDER RESIN

Title

Smart Bicycle Protective Helmet with Self-Healing Capabilities using Diels Alder Resin

Background of the Invention

Conventional helmets are designed to absorb impact, but they are typically single use after a significant impact. Users often discard helmets even after minor damage due to a lack of information on the helmet's remaining integrity. There is also no mechanism for recovery from superficial or moderate damage. Recent advancements in self-healing materials and embedded sensors open the possibility of smarter, more sustainable protective gear.

Summary of the Invention

The invention relates to a smart helmet system comprising: A multilayer protective shell with a self-healing outer layer, a localized healing trigger mechanism, activated post impact, based on sensor data. This system enables partial or complete regeneration of helmet components, improving helmet longevity, safety, and sustainability.

US20180121155A1 – Helmet for Smart Cycling Equipment

A helmet for smart cycling equipment includes a helmet body, a control unit, a speaker, a microphone, and a battery. The control unit is configured to wirelessly connect to a mobile device, enabling hands-free communication. The speaker and microphone facilitate audio input and output, while the battery powers the electronic components. This design aims to enhance safety by allowing cyclists to communicate without using handheld devices.

US20150305426A1 – Bicycle Helmet with Integrated Electronics

A bicycle helmet includes integrated electronics such as sensors, communication systems, and status indicators. These components monitor environmental conditions and relay information to a mobile device, enhancing rider awareness and safety through real-time data and alerts.

US6798392B2 – Smart Helmet

The smart helmet incorporates environmental monitoring systems and information delivery mechanisms to the user. It enhances safety by providing real-time data, such as location and environmental conditions, to improve situational awareness.

The proposed invention Smart Bicycle Protective Helmet with Self-Healing Capabilities using Diels Alder Resin advances significantly beyond existing technologies by combining multiple functionalities into a cohesive, intelligent system. This invention uniquely incorporates self-healing materials, such as Diels–Alder polymers to autonomously repair minor to moderate damage post-impact. The helmet features embedded sensors that trigger localized self-healing through a conductive mesh or printed ink heating system, all without user intervention. This automated healing not only enhances safety but also promotes sustainability by extending the helmet's usable lifespan features not concurrently addressed in any existing patent.

Detailed Description

Helmet Structure (100) comprises of outer shell (106) which is made from a self-healing polymer composite, such as a grapheme reinforced diels alder resin and micro capsule infused thermoplastic. Middle Impact Layer (101) consists of traditional EPS or EPP foam, optionally infused with viscoelastic gel for multi impact absorption. Inner Comfort Layer (109) has soft foam with moisture wicking and antibacterial properties. Impact Detection System incorporates flexible printed sensors (e.g., piezoelectric, MEMS accelerometers) embedded between the outer and middle layers. Sensors detect impact and activate the self-healing mechanism. Self-healing Activation Mechanism is for localized healing, the system uses printed conductive ink or resistive heating mesh in selected areas. When an impact is logged, the microcontroller triggers localized heating,

activating the healing process in the polymer. Power Supply helps small rechargeable battery powered by USB-C port or Optional flexible solar panels on the outer shell.

Brief Description of the Drawing

Figure 1A Process Flow Diagram illustrates the functional sequence of the smart helmet system (100). It begins with the detection (101) of an impact event using embedded sensors. The controller then triggers localized heating through conductive elements embedded in the shell. Finally, the self-healing (103) material responds to the heat, initiating the repair process. Sensors such as piezoelectric elements or MEMS accelerometers detect the force of an impact. The system compares the impact magnitude to pre-set safety thresholds (e.g., G-force values). This determines whether healing should be triggered or the helmet should be retired. If the damage is within repairable limits, the system activates localized conductive heating elements under the damaged area. The heat activates the self-healing (105) polymer layer, causing cracks or deformation to close or re-bond.

Figure 1B Section View of the Helmet This technical line drawing shows a front and side view of the helmet (100), highlighting the placement of embedded sensors (106), the location of impact-logging electronics, conductive mesh zones (107), and ventilation panels. The design balances aesthetics with embedded smart functionality. It is formed from Diels–Alder polymer composite (108), capable of healing scratches, dents, and shallow cracks. Hidden within the outer shell in zones prone to damage; activates localized heat when needed. Strategically placed around the shell to provide full coverage of head regions and detect impact vectors.

Self-Healing Shell (108) which is made of a thermally responsive Diels–Alder polymer composite. Materials used are Grapheme-reinforced Diels–Alder thermoset polymer. Functionality of Absorbs impact energy and autonomously heals surface damage using heat or pressure. Embedded Conductive Mesh Layer (107) is responsible for localized heating upon sensor activation. Materials used are Printed conductive ink and flexible metal mesh. Functionality of Triggers thermal activation of the healing process when damage is detected. Embedded System with Controller Layer (106) includes impact sensors, microcontroller and memory unit. Materials used are Flexible PCB, integrated with piezoelectric/MEMS sensors. Functionality is to Detect and analyses impacts; triggers heating system. Shock Absorbing Layer / Inner Comfort Layer (109) is made of viscoelastic foam or EPP, providing multi-impact resistance. It functions to Dissipates energy during a crash, prevents penetration to the skull, and regains shape after light impacts. Inner Comfort Layer is composed of breathable, antimicrobial padding for wearer comfort. It provides a function to ergonomically fit and wearer comfort; may include sweat-wicking and thermal regulation.

Claims

1. A protective helmet (100) comprising a self-healing Diels–Alder Polymer (108) beneath the outer shell.
2. The helmet (100) of claim 1, wherein the impact detection system includes piezoelectric or MEMS based force sensors (106).
3. The helmet (100) of claim 1, wherein the healing system comprises a conductive mesh (107) activated to generate localized heat.
4. The helmet (100) of claim 1, wherein an embedded impact detection system (106) with a controller configured to trigger localized repair based on detected impact.

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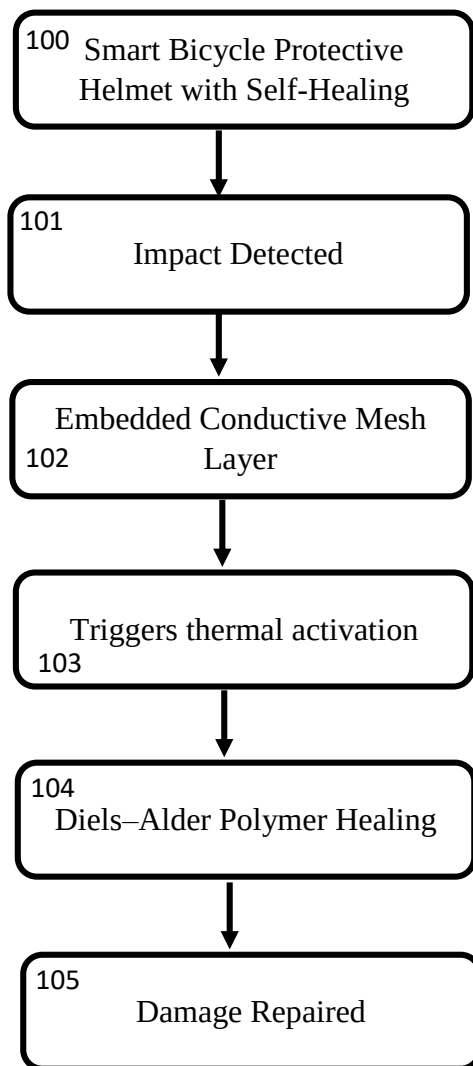


Figure 1A

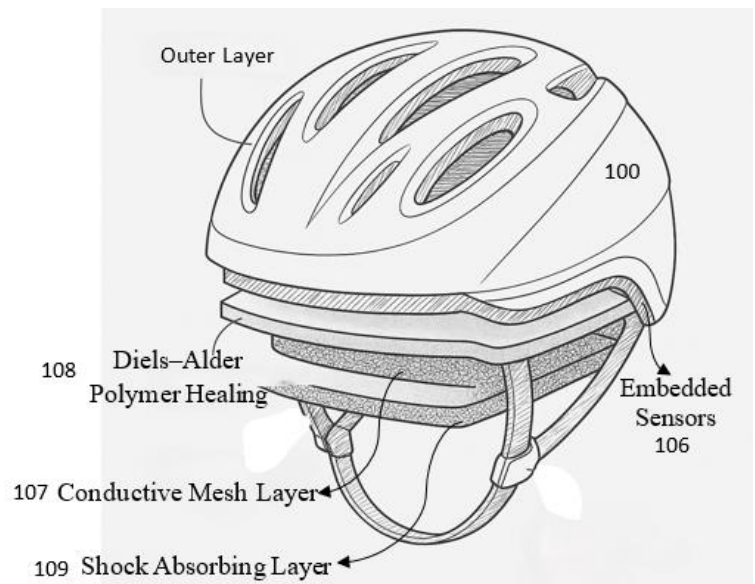


Figure 1B

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Abstract

A smart protective helmet (100) that integrates impact detection (101) and self-healing material (104) systems. Which is made of Diels–Alder Polymer (104). This invention extends the lifecycle of helmets (100) and enables automated recovery of structural and cosmetic damage via heat activated Diels–Alder Polymer (104) Healing polymers.

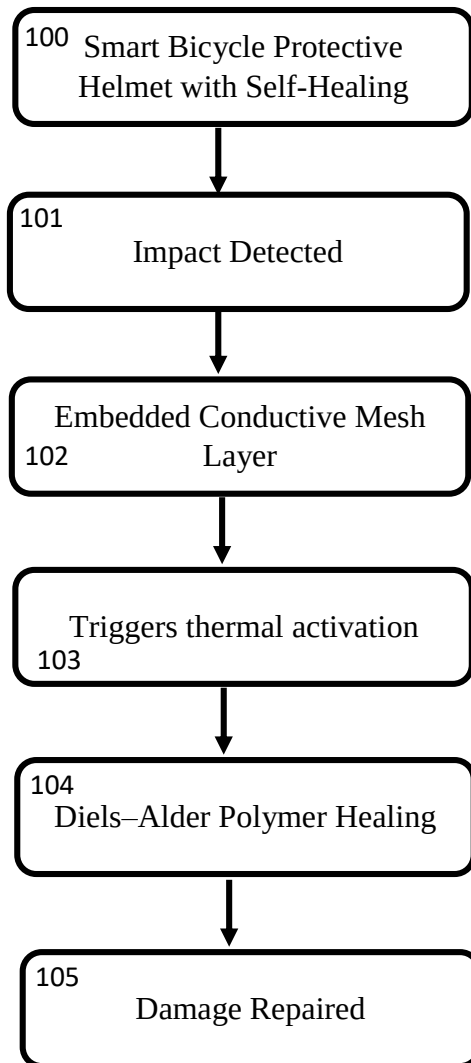


Figure 1A

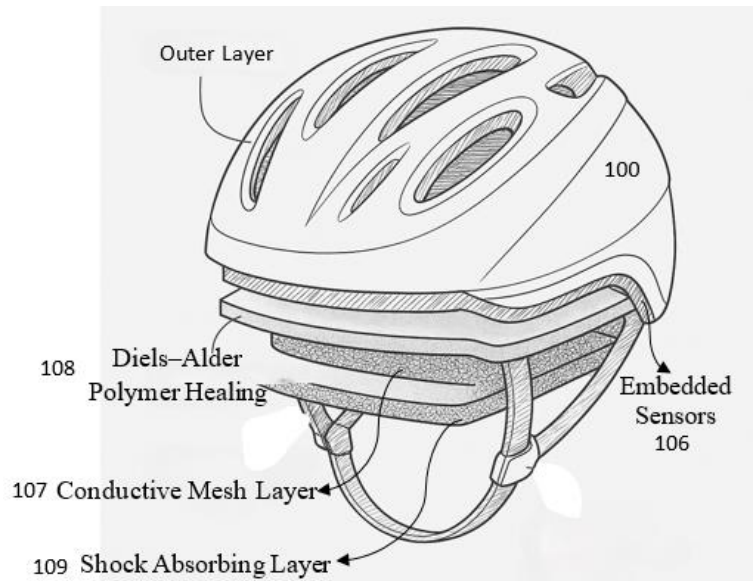


Figure 1B

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